Lab Number# 1

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3.Specific Requirements

3.1.Functional Requirements

3.1.1 User Registration and Authentication

- FR-01 — User registration: New users can create a BatchHub account by providing required information such as name, email address, student ID, batch, department, and password. The application validates the submitted information before creating the account.
- FR-02 — User login: Registered users can access their accounts by providing valid login credentials.
- FR-03 — Authentication validation: Invalid or incorrect credentials are rejected, and the user is informed that authentication was unsuccessful.
- FR-04 — User logout: Authenticated users can securely log out of their BatchHub account.
- FR-05 — Registration validation: Required registration fields are validated before account creation, and invalid or incomplete information is rejected with an appropriate message.

3.1.2 User Profile Management

- FR-06 — Profile viewing: Users can view their personal and academic profile information.
- FR-07 — Profile updating: Users can update permitted profile information such as name, profile picture, contact information, and other editable details.
- FR-08 — Profile access control: Users can modify only the profile information they are authorized to change.

3.1.3 Role-Based Access Control

- FR-09 — User roles: BatchHub supports different user roles, such as students, authorized academic users, and administrators, with different levels of access.
- FR-10 — Permission management: Access to management functions is determined by the user's assigned role and permissions.
- FR-11 — Administrative management: Administrators can manage users and system-level academic information according to their assigned permissions.

3.1.4 Course Management

- FR-12 — Course creation: Authorized users can create course records containing information such as course title, course code, description, department, and batch.
- FR-13 — Course viewing: Students can view courses relevant to their batch or academic program.
- FR-14 — Course updating: Authorized users can modify existing course information.
- FR-15 — Course deletion: Authorized users can remove course records they are permitted to manage.
- FR-16 — Course organization: Academic resources such as notices and course materials are associated with their respective courses to keep academic information organized.

3.1.5 Notice Management

- FR-17 — Notice creation: Authorized users can create and publish academic notices containing information such as title, description, category, publication date, and target audience.

- FR-18 — Notice viewing: Students can view notices relevant to their batch, department, courses, or other authorized groups.
- FR-19 — Notice updating: Authorized users can edit notices they are permitted to manage.
- FR-20 — Notice deletion: Authorized users can remove notices they are permitted to manage.
- FR-21 — Notice categorization: Notices can be categorized into relevant categories such as academic, examination, event, or general announcements.

3.1.6 Course Material Management

- FR-22 — Material upload: Authorized users can upload course-related materials such as lecture notes, documents, and other supported files.
- FR-23 — Material information: Each uploaded material contains relevant metadata such as title, course, category, uploader, and upload date.
- FR-24 — Material viewing: Students can view course materials associated with courses they are authorized to access.
- FR-25 — Material download: Students can download course materials that are available to them.
- FR-26 — Material management: Authorized users can update or remove course materials they are permitted to manage.
- FR-27 — Material organization: Course materials are organized by course and category to make academic resources easier to find.

3.1.7 Project and Group Management

- FR-28 — Project creation: Students can create academic project records containing information such as project title, description, course or academic context, and project members.
- FR-29 — Project viewing: Users can view project information that is available to them.
- FR-30 — Group creation: Students can create groups for academic projects.
- FR-31 — Group joining: Students can join available project groups according to the group's membership rules.
- FR-32 — Group member management: Authorized group members can view and manage group membership according to their permissions.
- FR-33 — Project updating: Authorized project members can update relevant project information.
- FR-34 — Project deletion: Authorized users can remove projects they are permitted to manage.

3.1.8 Search and Filtering

- FR-35 — Academic search: Users can search for relevant academic information, including courses, notices, course materials, and projects.
- FR-36 — Keyword search: Users can search using keywords such as course name, course code, notice title, material title, or project title.
- FR-37 — Information filtering: Search results can be filtered according to relevant attributes such as course, category, batch, or date.
- FR-38 — Permission-aware results: Search results contain only information that the current user is authorized to access.

3.1.9 Notifications

- FR-39 — Notification generation: BatchHub generates notifications when important events occur that are relevant to a user or group.

- FR-40 — Academic notifications: Users receive relevant notifications for events such as newly published notices, newly available course materials, and project or group updates.
- FR-41 — Notification viewing: Users can view their current and previously received notifications.
- FR-42 — Read/unread status: Notifications are identified as read or unread, allowing users to distinguish new updates from previously viewed ones.
- FR-43 — Notification targeting: Notifications are delivered only to users or groups relevant to the corresponding event.

3.2 Non-Functional Requirements

3.2.1 Performance

- NFR-01 — Application response time: Common operations such as loading the dashboard, viewing a course, opening a notice, or retrieving a material should complete within an acceptable response time under normal operating conditions.
- NFR-02 — Search response: Search and filtering operations should return results without noticeable delay under the expected normal user load.

3.2.2 Security

- NFR-03 — Secure authentication: User authentication credentials are handled using appropriate security mechanisms to prevent unauthorized account access.
- NFR-04 — Password protection: User passwords are stored using a secure password-hashing mechanism and are never stored as plain text.
- NFR-05 — Access control: Users cannot access or modify resources outside their assigned permissions.
- NFR-06 — Secure communication: Client-server communication uses HTTPS to protect credentials, personal information, and other sensitive data during transmission.

3.2.3 Usability

- NFR-07 — User-friendly interface: The application provides an intuitive interface that allows students and authorized users to access common features without requiring specialized technical knowledge.
- NFR-08 — Consistent navigation: Similar functions use consistent navigation, terminology, and interaction patterns throughout the application.
- NFR-09 — Error feedback: Invalid input, failed operations, and system errors are presented using understandable messages rather than raw technical error information.

3.2.4 Responsiveness and Compatibility

- NFR-10 — Responsive interface: The application provides a usable interface across commonly used desktop, tablet, and mobile screen sizes.
- NFR-11 — Browser compatibility: BatchHub supports commonly used modern web browsers.

3.2.5 Reliability

- NFR-12 — Error handling: Application errors are handled gracefully without exposing sensitive system information to users.

- NFR-13 — Data consistency: Related user, course, notice, material, project, group, and notification data remain consistent after create, update, and delete operations.
- NFR-14 — Backup and recovery: Persistent system data can be backed up and recovered in the event of system failure or data loss.

3.2.6 Scalability

- NFR-15 — User scalability: The application is designed to support an increasing number of students, academic users, courses, notices, materials, and projects.
- NFR-16 — Resource scalability: The application architecture can be expanded with additional server, database, and storage resources as usage increases.

3.2.7 Maintainability

- NFR-17 — Modular design: Application components are organized into modules so that individual features can be modified or extended without unnecessarily affecting unrelated functionality.
- NFR-18 — Code maintainability: The project follows consistent coding practices, meaningful naming conventions, and appropriate documentation to make future maintenance easier.

3.2.8 Privacy

- NFR-19 — Personal information protection: Personal and academic information is accessible only to users with appropriate permissions.
- NFR-20 — Data minimization: The application collects and stores only the information required to provide its intended functionality.

3.3 External Interface Requirements

3.3.1 User Interfaces

- Web-based interface: BatchHub provides a browser-based interface through which students, authorized academic users, and administrators access the platform.
- Authentication interface: Registration, login, and logout interfaces provide users with access to their accounts.
- Dashboard: The dashboard provides users with an overview of relevant academic information, recent notices, materials, projects, and notifications.
- Course interface: Users can browse courses and access course-related information and materials according to their permissions.
- Notice interface: Users can view relevant notices, while authorized users can create, edit, categorize, and remove notices.
- Course material interface: Authorized users can upload and manage materials, while students can browse and download available resources.
- Project and group interface: Students can create projects, create or join groups, and manage project-related information according to their permissions.
- Search interface: Users can search and filter academic information using keywords and relevant filtering options.
- Notification interface: Users can view new and previously received notifications and distinguish between read and unread notifications.

3.3.2 Hardware Interfaces

- Client devices: No specialized hardware is required. Users can access BatchHub using standard desktop computers, laptops, tablets, or smartphones.
- Server infrastructure: The backend application, database, and file-storage services run on appropriate server or cloud infrastructure capable of supporting the expected workload.

3.3.3 Software Interfaces

- Frontend-backend interface: The frontend communicates with the backend through defined APIs for authentication, user management, courses, notices, materials, projects, groups, search, and notifications.
- Database interface: The backend communicates with the database to store and retrieve user, course, notice, material, project, group, and notification information.
- File storage interface: The backend communicates with the selected file-storage mechanism to store and retrieve uploaded course materials.
- Authentication interface: The authentication component communicates with the backend to verify user credentials and maintain authenticated sessions.

3.3.4 Communication Interfaces

- Client-server communication: Communication between the frontend and backend takes place through HTTP-based APIs.
- Secure communication: HTTPS is used for communication involving authentication credentials, personal information, academic information, and other sensitive data.
- API communication: API requests and responses use a structured data format such as JSON.
- Protected API access: Requests to protected resources include appropriate authentication information, and the backend validates the user's permissions before processing the request.